

Project Report: Real-Time Streaming Pipeline - Data Engineer

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Family:	Data Engineer
Level:	Advanced
Chapters:	6
Source:	CodeQuest project specification (official docs + industry patterns)

Summary

Kafka ? Spark Structured Streaming ? Delta Lake with monitoring and exactly-once semantics.

Chapters

Track: Project-Intro

Chapter 1: Project Overview [Advanced]

Chapter focus: Project Overview** *(Level: Advanced)

Real-Time Streaming Pipeline** is a advanced-level end-to-end project for Data Engineer. **Duration:** 6-8 weeks. **Kafka ? Spark Structured Streaming ? Delta Lake** with monitoring and exactly-once semantics. **Tools:** Kafka, Spark, Delta Lake, Grafana.

Code Reference:

```
Objective: Kafka topic ingestion
Objective: Spark streaming transforms
Objective: Delta Lake sink
Objective: Grafana monitoring dashboard
```

What it shows: Lists measurable outcomes before you start building.

Topic guide

What it actually is

A project overview defines scope, tools, and success criteria.

When to use it

At project kickoff.

Where to use it

Data Engineer portfolios and CodeQuest project submissions.

Real use example

A candidate lists this project on a Data Engineer resume with a demo link.

Track: Project-Phase

Chapter 2: Phase: Kafka [Advanced]

Chapter focus: Phase: Kafka** *(Level: Advanced)

Producer + consumer. Complete these tasks: Avro schema; Partitions. Deliverable for this phase: Topic flowing.

Code Reference:

- [] Avro schema
- [] Partitions

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Kafka is one step in the full project lifecycle.

When to use it

During week 1 of the project timeline.

Where to use it

Data Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Topic flowing to the teacher.

Chapter 3: Phase: Spark [Advanced]

Chapter focus: Phase: Spark** *(Level: Advanced)

Structured streaming job. Complete these tasks: Window aggregations; Watermark. Deliverable for this phase: Streaming job running.

Code Reference:

- [] Window aggregations
- [] Watermark

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Spark is one step in the full project lifecycle.

When to use it

During week 2 of the project timeline.

Where to use it

Data Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Streaming job running to the teacher.

Chapter 4: Phase: Delta [Advanced]**Chapter focus: Phase: Delta** *(Level: Advanced)**

ACID sink. Complete these tasks: Merge upserts; Time travel query. Deliverable for this phase: Delta table.

Code Reference:

- [] Merge upserts
- [] Time travel query

What it shows: Checklist format helps track phase completion.

Topic guide**What it actually is**

Phase Delta is one step in the full project lifecycle.

When to use it

During week 3 of the project timeline.

Where to use it

Data Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Delta table to the teacher.

Chapter 5: Phase: Monitor [Advanced]**Chapter focus: Phase: Monitor** *(Level: Advanced)**

Metrics dashboard. Complete these tasks: Lag; Throughput; Errors. Deliverable for this phase: Grafana dashboard.

Code Reference:

- [] Lag
- [] Throughput
- [] Errors

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Monitor is one step in the full project lifecycle.

When to use it

During week 4 of the project timeline.

Where to use it

Data Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Grafana dashboard to the teacher.

Track: Project-Deliverables

Chapter 6: Final Deliverables and Review [Advanced]

Chapter focus: Final Deliverables and Review *(Level: Advanced)**

Submit all final deliverables: Streaming job code; Delta table; Grafana dashboard; Architecture doc. Skills demonstrated: Kafka, Spark, Streaming. Prepare a 5-minute walkthrough video and README for CodeQuest teacher marking.

Code Reference:

Deliverable: Streaming job code
Deliverable: Delta table
Deliverable: Grafana dashboard
Deliverable: Architecture doc

What it shows: Final checklist ensures nothing is missing at submission.

Topic guide

What it actually is

Deliverable review confirms end-to-end project completion.

When to use it

Before deadline and teacher review.

Where to use it

CodeQuest project gallery and interviews.

Real use example

A student submits Streaming job code and passes the rubric.